Plane Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Angle relationships**

Two angles are supplementary, and one is 3 times the other. Find both angles.

Let the angles be x and $3x$: $x + 3x = 180$, so $4x = 180$, giving $x = 45$. The angles are 45° and 135°.

2 **[Foundational]** • **Target: Angle relationships**

Two parallel lines are cut by a transversal, creating an angle of 70°. Find the measure of its alternate interior angle and its corresponding angle.

Alternate interior angles are equal: 70°. Corresponding angles are also equal: 70°.

3 **[Foundational]** • **Target: Triangle angle sum and exterior angles**

A triangle has angles measuring x, $2x$, and $3x$. Find each angle.

$x + 2x + 3x = 180$, so $6x = 180$, giving $x = 30$. Angles: 30°, 60°, 90°.

4 **[Foundational]** • **Target: Triangle angle sum and exterior angles**

A triangle's exterior angle at one vertex measures 115°. If one of the remote interior angles is 50°, find the other remote interior angle.

By the exterior angle theorem, the exterior angle equals the sum of the two remote interior angles: $115 = 50 + x$, so $x = 65^\circ$.

5 **[Foundational]** • **Target: Area and perimeter of polygons**

Find the area of a triangle with base 14 and height 9.

Area $= \frac{1}{2}(14)(9) = 63$.

6 **[Foundational]** • **Target: Area and perimeter of polygons**

Find the area of a parallelogram with base 12 and height 5.

Area $= \text{base} \times \text{height} = 12 \times 5 = 60$.

7 **[Foundational]** • **Target: Area and perimeter of polygons**

Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.

$A = \frac{3\sqrt{3}}{2}(6)^2 = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The Pythagorean theorem and special right triangles**

A 45-45-90 triangle has a leg of length 8. Find the length of the hypotenuse.

In a 45-45-90 triangle, the hypotenuse equals $\text{leg} \times \sqrt{2}$: $8\sqrt{2}$.

9 **[Intermediate]** • **Target: The Pythagorean theorem and special right triangles**

A 30-60-90 triangle has a hypotenuse of 20. Find the lengths of both legs.

In a 30-60-90 triangle, the shorter leg is $\frac{\text{hypotenuse}}{2}$ and the longer leg is $\frac{\text{hypotenuse}}{2} \times \sqrt{3}$: shorter leg $= 10$, longer leg $= 10\sqrt{3}$.

10 **[Intermediate]** • **Target: Circles: area, circumference, arcs, and sectors**

Find the area and circumference of a circle with diameter 14, in terms of π.

Radius $= 7$. Area $= \pi r^2 = 49\pi$. Circumference $= \pi d = 14\pi$.

11 **[Intermediate]** • **Target: Circles: area, circumference, arcs, and sectors**

Find the arc length of a 120° sector in a circle with radius 9, in terms of π.

Arc length $= \frac{120}{360} \times 2\pi(9) = \frac{1}{3} \times 18\pi = 6\pi$.

12 **[Intermediate]** • **Target: Similar and congruent figures**

Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.

Areas of similar figures scale by the SQUARE of the linear scale factor: $\left(\frac{5}{3}\right)^2 = \frac{25}{9}$. Larger triangle's area: $18 \times \frac{25}{9} = 50$.

13 **[Intermediate]** • **Target: Volume and surface area of 3D solids**

Find the volume of a rectangular prism with dimensions 5, 8, and 10.

Volume $= 5 \times 8 \times 10 = 400$.

14 **[Intermediate]** • **Target: Volume and surface area of 3D solids**

Find the volume of a cone with radius 6 and height 10, in terms of π.

Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(10) = 120\pi$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Multi-step geometry problems • Cross-Lesson: 6.01 Trigonometry

This question has two parts. A square garden has area 144 square feet. (a) Find the side length and perimeter of the garden. (b) A circular fountain with radius 3 feet is placed inside the garden. Find the remaining garden area (not covered by the fountain), in terms of π.

(a) Side length: $\sqrt{144} = 12$ feet. Perimeter: $4(12) = 48$ feet. (b) Fountain area: $\pi(3)^2 = 9\pi$. Remaining area: $144 - 9\pi$ square feet.

16 [Challenging] • Target: Trapezoids and composite polygon areas • Cross-Lesson: 7.01 Statistics & Probability

Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

Area $= \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66$.

17 [Challenging] • Target: Trapezoids and composite polygon areas • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

An L-shaped figure is formed by a 10-by-8 rectangle with a 4-by-3 rectangle removed from one corner. Find the remaining area.

Large rectangle area: $10 \times 8 = 80$. Removed rectangle area: $4 \times 3 = 12$. Remaining area: $80 - 12 = 68$.

18 [Challenging] • Target: Chords, tangents, and inscribed angles in circles • Cross-Lesson: 1.01 Numbers, Percents & Ratios

An inscribed angle subtends an arc of 100°. Find the measure of the inscribed angle.

An inscribed angle is always half the arc it subtends: $\frac{100^\circ}{2} = 50^\circ$.

19 [Challenging] • Target: Chords, tangents, and inscribed angles in circles • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A tangent line touches a circle at point P. Explain the relationship between the tangent line and the radius drawn to point P.

A tangent line is always perpendicular to the radius drawn to the point of tangency — this is a fundamental property relating tangent lines and circles.

20 [Challenging] • Target: Congruence and triangle proofs • Cross-Lesson: 3.01 Functions & Graphs

State the triangle congruence criterion (SSS, SAS, ASA, AAS) that applies when two triangles share two equal angles and the included side.

ASA (Angle-Side-Angle) applies, since two angles and the side directly between them are all given as congruent.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Angle relationships

Two angles are complementary, and they differ by 52°. Find the larger angle.

a. 19°

b. 66°

c. 76°

d. 71° ✓

Let the angles be x and $x + 52$: $x + (x + 52) = 90$, so $2x = 38$, giving $x = 19$. The larger angle is $19 + 52 = 71^\circ$.

22 [Foundational] • Target: Triangle angle sum and exterior angles

Two angles are supplementary, and they differ by 4°. Find the larger angle.

a. 87°

b. 88°

c. 97°

d. 92° ✓

Let the angles be x and $x + 4$: $x + (x + 4) = 180$, so $2x = 176$, giving $x = 88$. The larger angle is $88 + 4 = 92^\circ$.

23 [Foundational] • Target: Area and perimeter of polygons

A right triangle has legs 12 and 16. Find the length of the hypotenuse.

a. 28

b. 21

c. 19

d. 20 ✓

$c = \sqrt{12^2 + 16^2} = \sqrt{144 + 256} = \sqrt{400} = 20$.

24 [Foundational] • Target: The Pythagorean theorem and special right triangles

A circle has radius 11. Find the arc length of a 270° sector, in terms of π.

a. $\frac{33}{2}\pi$ ✓

b. $\frac{35}{2}\pi$

c. $\frac{31}{2}\pi$

d. 33π

Arc length $= \frac{\theta}{360^\circ} \times 2\pi r = \frac{270}{360} \times 2\pi(11) = \frac{33}{2}\pi$.

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25 [Foundational] • Target: Circles: area, circumference, arcs, and sectors

A circle has radius 2. Find the sector area of a 180° sector, in terms of π.

a. 2π ✓

b. 3π

c. 4π

d. 1π

Sector area $= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{180}{360} \times \pi(4) = 2\pi$.

26 [Foundational] • Target: Similar and congruent figures

Find the volume of a cylinder with radius 5 and height 14, in terms of π.

a. 350π ✓

b. 360π

c. 140π

d. 340π

Volume $= \pi r^2 h = \pi(25)(14) = 350\pi$.

27 [Foundational] • Target: Volume and surface area of 3D solids

In a triangle, two remote interior angles measure 50° and 50°. What is the exterior angle at the third vertex?

a. **100**$^\circ$ ✓

b. 45°

c. 53°

d. 178°

The exterior angle equals the sum of the two remote interior angles: $50 + 50 = 100$.

28 [Foundational] • Target: Multi-step geometry problems

Two similar triangles have a scale factor of 3. If the smaller triangle has area 9, what is the area of the larger triangle?

a. 136

b. 86

c. 147

d. **81** ✓

Area ratio $= k^2 = (3)^2$. Larger area $= 9 \times 9 = 81$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Trapezoids and composite polygon areas

A right circular cylinder has height 12 and base radius 5. A sphere is inscribed inside the cylinder such that it touches the cylinder's curved surface. What is the volume of the space inside the cylinder that is outside the sphere?

a. 100π

b. $\frac{400\pi}{3}$ ✓

c. $\frac{500\pi}{3}$

d. 200π

Sphere radius is $r = 5$. Cylinder volume $V_c = \pi r^2 h = \pi(25)(12) = 300\pi$. Sphere volume $V_s = \frac{4}{3}\pi(5^3) = \frac{500\pi}{3}$. Space outside sphere: $300\pi - \frac{500\pi}{3} = \frac{900\pi - 500\pi}{3} = \frac{400\pi}{3}$.

30 [Intermediate] • Target: Chords, tangents, and inscribed angles in circles

What is each exterior angle of a regular 12-sided polygon?

a. 54°

b. 42°

c. **30°** ✓

d. 4°

Exterior angles sum to 360°; each $= 360/12 = 30$.

31 [Intermediate] • Target: Congruence and triangle proofs

In a triangle, two of the angles measure 38° and 74°. What is the measure of the third angle?

a. 64°

b. **68°** ✓

c. 66°

d. 59°

Triangle angles sum to 180°. $180 - 38 - 74 = 68^\circ$

32 [Intermediate] • Target: A multi-part composite geometry problem

Two angles of a triangle measure 81° and 54°. What is the third angle?

a. 39°

b. **45°** ✓

c. 171°

d. 1°

Angles sum to 180°: $180 - 81 - 54 = 45$.

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33 [Intermediate] • Target: Angle relationships

What is the sum of the interior angles of a 5-sided polygon?

a. 879°

b. **540°** ✓

c. 430°

d. 75°

Sum $= (n-2) \times 180 = (5-2) \times 180 = 540$.

34 [Intermediate] • Target: Triangle angle sum and exterior angles

Two similar triangles have a scale factor of 5. If a side of the smaller triangle is 5, what is the corresponding side of the larger triangle?

a. 37

b. 36

c. **25** ✓

d. 19

Corresponding side $= 5 \times 5 = 25$.

35 [Intermediate] • Target: Area and perimeter of polygons

In a right triangle with acute angles A and B, $\sin(A) = \frac{5}{13}$. What is the value of $\cos(B) + \tan(A)$?

a. $\frac{5}{13} + \frac{5}{12} = \frac{125}{156}$ ✓

b. $\frac{12}{13} + \frac{5}{12}$

c. $\frac{5}{13} + \frac{12}{5}$

d. 1

Since A and B are complementary acute angles in a right triangle, $\cos(B) = \sin(A) = \frac{5}{13}$. With opposite = 5 and hypotenuse = 13, adjacent = 12, so $\tan(A) = \frac{5}{12}$. Then $\cos(B) + \tan(A) = \frac{5}{13} + \frac{5}{12} = \frac{60+65}{156} = \frac{125}{156}$.

36 [Intermediate] • Target: The Pythagorean theorem and special right triangles

A right triangle has area 24 and hypotenuse 10. What is the perimeter of the triangle?

a. 20

b. 22

c. **24** ✓

d. 26

Let legs be a, b. $\frac{1}{2}ab = 24 \Rightarrow ab = 48$. Also $a^2 + b^2 = 10^2 = 100$. Then $(a+b)^2 = a^2 + b^2 + 2ab = 100 + 96 = 196 \Rightarrow a + b = 14$. Perimeter $= a + b + c = 14 + 10 = 24$.

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37 [Intermediate] • Target: Circles: area, circumference, arcs, and sectors

A right triangle has hypotenuse 10 and one acute angle measuring 30°. What is the area of the triangle?

a. $12.5\sqrt{3}$ ✓

b. $25\sqrt{3}$

c. 25

d. 50

Opposite side (height) = $10 \times \sin(30^\circ) = 5$. Adjacent side (base) = $10 \times \cos(30^\circ) = 5\sqrt{3}$. Area = $\frac{1}{2}(5)(5\sqrt{3}) = 12.5\sqrt{3}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Similar and congruent figures • Cross-Lesson: 1.01 Numbers, Percents & Ratios

In a right triangle with acute angle θ, the side opposite θ is 9 and the hypotenuse is 15. What is $\sin(\theta)$?

a. $\frac{3}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{3}{4}$

d. $\frac{5}{3}$

$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{9}{15} = \frac{3}{5}$.

39 [Challenging] • Target: Volume and surface area of 3D solids • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

In a right triangle, acute angle A measures 30°. What is $\sin(30^\circ)$?

a. $\frac{1}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$

d. 1

From standard 30-60-90 right triangle ratios, $\sin(30^\circ) = \frac{1}{2}$.

40 [Challenging] • Target: Multi-step geometry problems • Cross-Lesson: 3.01 Functions & Graphs

In a right triangle, the side opposite angle θ has length 8, the side adjacent to θ has length 15, and the hypotenuse has length 17. What is $cos(\theta)$?

a. 77/68

b. 283/340

c. 92/85

d. **15/17** ✓

$cos(\theta) = 15/17$ using SOH-CAH-TOA with sides $8, 15, 17$.

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41 [Challenging] • Target: Trapezoids and composite polygon areas • Cross-Lesson: 4.01 Coordinate Geometry

In a right triangle, the length of the hypotenuse is 12, and the sum of the lengths of the two legs is 16. If θ is the measure of one of the acute angles of the triangle, what is the value of $\sin(2\theta)$?

a. $\frac{7}{18}$

b. $\frac{7}{16}$

c. $\frac{5}{9}$

d. $\frac{7}{9}$ ✓

Let the lengths of the two legs be a and b, and the hypotenuse be $c = 12$. From the problem statement: $a + b = 16$ By the Pythagorean theorem: $a^2 + b^2 = c^2 = 12^2 = 144$ We can find the product ab by squaring both sides of the sum equation: $(a + b)^2 = a^2 + 2ab + b^2$ $16^2 = (a^2 + b^2) + 2ab$ $256 = 144 + 2ab \Rightarrow 2ab = 112 \Rightarrow ab = 56$ For an acute angle θ in a right triangle, the trigonometric ratios of the two legs relative to the hypotenuse are $\sin\theta = \frac{a}{c}$ an.

42 [Challenging] • Target: Chords, tangents, and inscribed angles in circles • Cross-Lesson: 6.01 Trigonometry

An equilateral triangle has sides of length 10. What is its perimeter?

a. 20

b. 25

c. **30** ✓

d. 35

Applying Area, Volume, Triangles, Circles & Trigonometry principles, Perimeter = $3 \times 10 = 30$. Therefore, the verified correct option is 30.

43 [Challenging] • Target: Congruence and triangle proofs • Cross-Lesson: 7.01 Statistics & Probability

A right triangle has legs of length 5 and 12. What is the length of the hypotenuse?

a. **13** ✓

b. 15

c. 17

d. $\sqrt{119}$

Applying Area, Volume, Triangles, Circles & Trigonometry principles, $\sqrt{25 + 144} = \sqrt{169} = 13$. Therefore, the verified correct option is 13.

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44 [Challenging] • Target: A multi-part composite geometry problem • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

In a right triangle, the legs are 5 and 12. What is the hypotenuse?

a. **13 ✓**

b. 15

c. 17

d. $\sqrt{119}$

Applying Area, Volume, Triangles, Circles & Trigonometry principles, $\sqrt{25+144} = \sqrt{169} = 13$

Therefore, the verified correct option is 13.

45 [Challenging] • Target: Angle relationships • Cross-Lesson: 1.01 Numbers, Percents & Ratios

If two angles in a triangle are 45° and 45°, what type of triangle is it?

a. Acute equilateral

b. **Right isosceles ✓**

c. Obtuse isosceles

d. Right scalene

The third angle is $180^\circ - 45^\circ - 45^\circ = 90^\circ$. Two equal angles make it isosceles, and a 90° angle makes it a right triangle.

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